Veronica Serrano

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Education

Master of Science, Geological Sciences Expected May 2027

The University of Texas at El Paso
Thesis: *Semi-Automated Geologic Mapping Using Machine Learning*
Advisor: Dr. Jose Miguel Hurtado Jr.

Bachelor of Science, Geological Sciences May 2025

The University of Texas at El Paso

Research Experience & Training

Master's Thesis Research — Selenograph August 2025–present

The University of Texas at El Paso

- Developed Selenograph, a Python application for semi-automated geologic mapping of lunar south polar terrain, integrating expert annotation, machine-learning predictions, and georeferenced GIS exports.
- Implemented Random Forest, XGBoost, and U-Net approaches for geologic-unit mapping to support comparison of tree-based and deep-learning methods.
- Built a geospatial processing workflow spanning 237 lunar terrain tiles across 13 sites at native 5 m resolution, with six terrain features derived from digital elevation models.
- Created map-level held-out evaluation tools reporting precision, recall, F1 score, intersection over union, and confusion matrices against human annotations.

Research Intern June–August 2025

Texas A&M AgriLife Research Center at El Paso | IMAGEN

- Analyzed 25 years of MODIS observations (2000–2024) to investigate vegetation and surface-water dynamics in the Rio Grande River Basin using Google Earth Engine, Python, and Jupyter Notebook.
- Generated annual NDVI and NDWI maps and time series with cloud filtering; compared land-cover maps from 2001 and 2023 to characterize landscape change.
- Produced a coauthored research poster documenting drought-related vegetation and water-index variability and changes in woody savannas, evergreen forests, and cropland.

Analog Astronaut — ARDVARC May 2025

Augmented Reality Data Visualization Analog Research Campaign

- Tested with and without augmented-reality support as an analog astronaut in a NASA PSTAR-funded campaign evaluating the use of rover-collected terrain data for lunar navigation.

Teaching Experience

Graduate Teaching Assistant August 2025–present

The University of Texas at El Paso

- Teach laboratory courses in Principles for Earth Sciences, Earth Materials, and Historical Geology.
- Prepare instructional materials for geology laboratory courses.

Technical Skills

Scientific computing: Python; NumPy; SciPy; scikit-learn; Jupyter Notebook
Machine learning: Random Forest; XGBoost; U-Net; classification and segmentation metrics
Geospatial tools: ArcGIS; QGIS; Rasterio; Google Earth Engine; ENVI Classic
Methods: Terrain analysis; geologic mapping; NDVI/NDWI time series; GeoTIFF processing
Research software: Streamlit; interactive annotation; reproducible geospatial workflows

Research Outputs & Conference Participation

Geological Society of America Connects October 2026

Serrano, V., Hurtado, M. J. (2026). *Can semi-automated learning methods standardize lunar geologic mapping and improve the efficiency of exploration site characterization?* Research poster.

UTEP Colloquium April 2026

Serrano, V., Hurtado, J., Sweeney, T. (2026). *Simulating Artemis Lunar Observations: A student -based mission exercise.* Research Poster

UTEP COURI Symposium August 2025

Serrano, V., Dayal, D., & Palmate, S. S. (2025). *Spatiotemporal Analysis of Vegetation and Water Dynamics in the Rio Grande River Basin using Google Earth Engine.* Research poster.

UTEP Colloquium April 2025

Serrano, V., Ortiz, A., Silva, J., Castillo, R., (2025). *Observing subsurface data using geophysical methods at the Jornada Experimental Range and Tornillo Pecan Orchard.* Research poster

Certifications & Additional Training

GeoAI–GIST Certificate August 2025–May 2027

Geospatial Information Science and Technology and Geospatial Artificial Intelligence

Outreach & Community Engagement

UTEP Pathways Summer Camp July 2026

The University of Texas at El Paso | Camp Leader

- Developed and led earth science activities and field trips for 30 high schoolers.

TRIO Upward Bound June 2026

New Mexico State University | Camp Leader

- Developed and led a camp that introduced high school students to Earth science topics.

Frontera Land Alliance / Cielo-G May 2024–May 2025

The University of Texas at El Paso | Intern

- Created an educational program that focused on land conservation, earth science, and community outreach through guided hikes and school events.

Leadership & Professional Memberships

UTEP Miners Planetary Council (UMPC) — Secretary 2026–present

Geological Society of America (GSA) — Student Member 2026–present

American Geophysical Union (AGU) — Student Member 2025–present

Association of Women in Geoscience (AWG) — Member 2024–present

Society of Advancements for Chicanos and Natives Americans in Science (SACNAS) — Member 2024–present